

ROGUE — Research Brief

A solo research build of a continuous open-web LLM red-team. The methods and the measured results — including the negative ones.

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FINDING 01 — CALIBRATING AN LLM-AS-JUDGE AGAINST HUMAN LABELS, THEN RECALIBRATING WHEN A BENCHMARK EXPOSED IT

Every breach verdict is an LLM judgment, so the judge is the load-bearing weakness. It was validated four ways — three against independent human-annotated benchmarks: blind stratified in-distribution hand-labels, WildGuardTest (Allen AI annotators), StrongREJECT, and JailbreakBench's *judge_comparison* (300 human-labeled rows against four field classifiers). JBB exposed over-flagging: the v1 judge agreed with the human majority only **70.3% — last of five** (behind HarmBench / LlamaGuard-2 / GPT-4 / Llama-3), at recall 98% / precision 55%.

A 20-row false-positive audit diagnosed **five recurring failure modes**; the root cause was a *rubric* problem — it rewarded engagement with the attack frame (persona acceptance, acknowledgment, format mimicry) over transfer of harmful content. A **content-transfer-gate rubric (v3)** moved the same 300 rows to **89.3% agreement / 79.5% precision / 95.5% recall** (+19 / +24.5 / −2.5 pp) — dead-last to 3rd of five, tied with the frontier classifiers — for ~\$8.4 via a tiered evaluation. Then the honest part: re-judging the stored breach matrix under v3 **dropped breach cells 2,429 to 1,371 (−43.6%)**, correcting prior over-reporting (all three external axes re-measured under v3: WildGuard harm 88.5%, StrongREJECT inflation −26%).

Why this is notable. A named false-positive taxonomy for a safety judge, plus a measured finding that two respected benchmarks (WildGuardTest harm labels, StrongREJECT) themselves over-count relative to a strict content-transfer standard.

FINDING 02 — SCHEDULING AS A CAPABILITY LEVER, NOT JUST AN OPTIMIZATION

A within-tier greedy reorder was replaced with a **target-conditioned cross-tier scheduler** — a static, explainable blend (0.5·global + 0.3·vendor + 0.2·family breach-rate, Laplace-smoothed, deliberately no ML / no bandit so it stays reproducible). A single-variable controlled experiment (same ladder, attacks, corpus, judge, and target — Claude Haiku, AdvBench + JailbreakBench; only the order changed) beat the production baseline on every axis: **median winner-rank 22 to 11–13.5, attack-success-rate 50% to 60%, and cost-per-success \$1.25 to \$0.74 (−41%)**.

Why this is notable. The mechanism: rank↓ caused ASR↑ — the old order exhausted the per-scan budget cap before reaching the winning technique — so reordering improved coverage, cost, and latency at once with zero new attacks. The reproducibility invariant is “reorder, never exclude”: same ladder, different order, full reachability preserved.

FINDING 03 — A PUBLICATION-GRADE NULL RESULT: GRAMMAR-COMPONENT PREDICTIVE POWER

Before building a grammar/AST attack-composition engine, a **\$0 observational study over 1,540 (primitive × target) cells** tested whether grammar-structure nodes predict breach *beyond* attack-family membership, with full confound controls: Benjamini–Hochberg FDR across hundreds of node/pair tests, Mantel–Haenszel stratification by target model, within-family lift, and Cramér's-V collinearity flagging. Verdict **weak/none** — the family label carries the predictive weight; cross-family structural nodes show ~1.0–1.1× non-significant lift, and the striking pre-FDR pairwise synergies (odds ratios up to 16.8) survived **none** of the four controls.

Why this is notable. A cheap, rigorous falsification that redirected engineering away from a months-long build — a successful negative result.

FINDING 04 — MEASURE-BEFORE-BUILD DISCIPLINE

\$0 measurements from existing telemetry were used repeatedly to *invert* “build it” decisions, each parked with an explicit trigger-to-revisit:

- **Per-model ladder routing** — the spread was a model main effect, not a family×model interaction, so it was not worth the rewrite.
 - **LLM renderer-synthesis** — synthesis-grade backlog flat at 7 across two widening harvests, so it was parked.
 - **HF jailbreak-dataset bulk-import** — measured 0 new attack families, so it was declined.
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LIMITATIONS (STATED PLAINLY)

Targets are black-box live-API models whose versions are not pinned; some cells are small-n (95% bootstrap CIs are persisted precisely because of this); the judge is single-operator-calibrated. These are descriptive measurements of a live system, not validated generalizations.